Admin

Assign 2

Tue 5pm for on-time bonus

Lab 3

Uart, C-strings, gdb

Today: Modules, Linking

Modules and libraries

- What makes for good design?

Build process

- How do source files become an executable?
- What does the linker do?
- Understanding and diagnosing build errors

Start sequence

- How does a program begin executing?

Road map

Instruction set architecture

Peripherals: GPIO, timers, UART

Assembly language and machine code

From C to assembly language

Function calls and stack frames

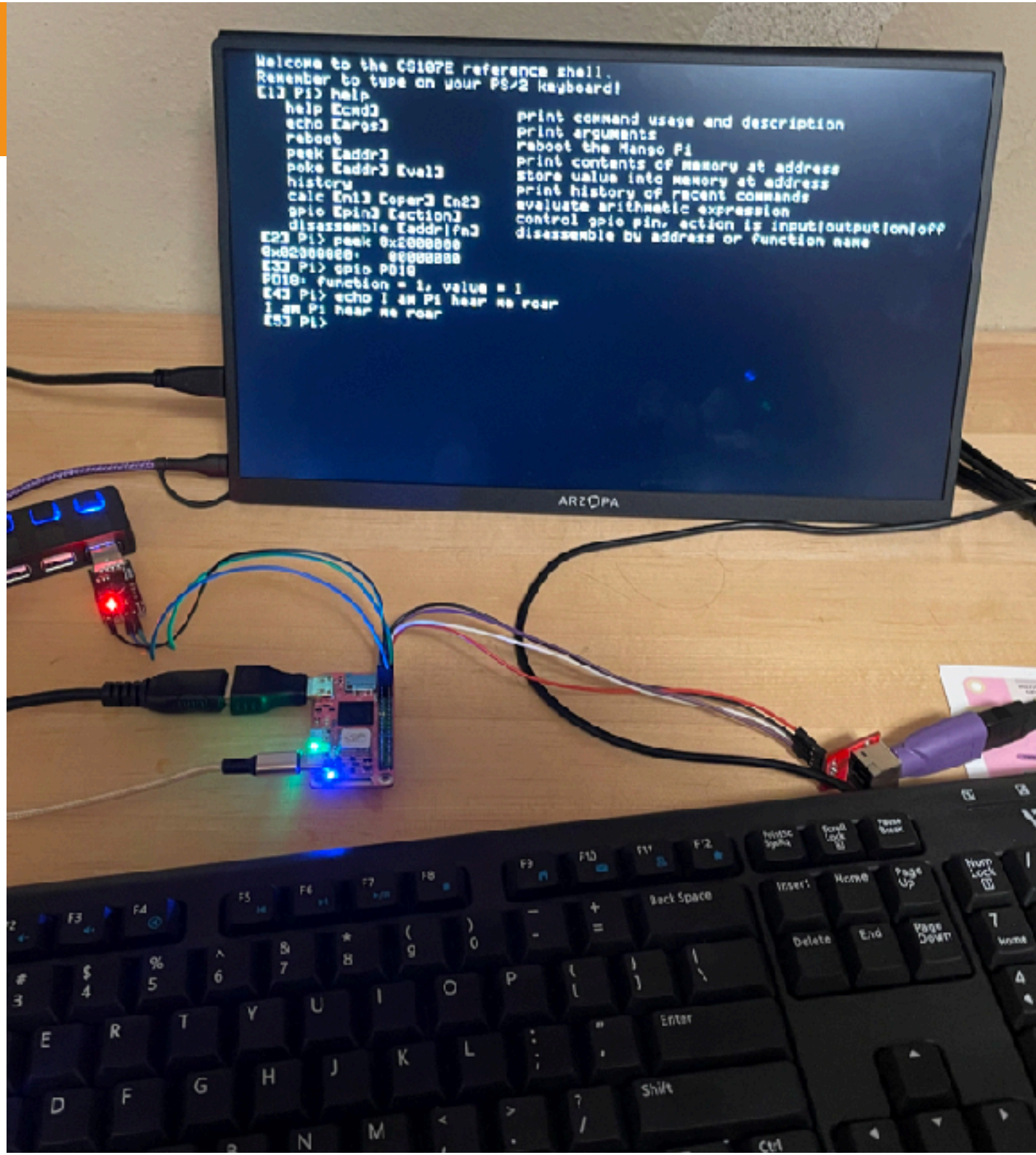
Serial communication and strings

➜ **Modules and libraries: Building and linking**

Memory map, stack & heap

libmango

gpio ✓
timer ✓
uart ✓
strings
printf
malloc
keyboard
shell
fb
gl
console



Good software design

Decompose a system into smaller parts (modules)

- **Interface**: what the module does
- **Implementation**: how it does it

A good interface:

- Easy-to-understand abstraction that simplifies client use
- Clear spec, portable, can swap implementations

Testing contract, module unit tests

Designing good interface boundaries is the "art" of software engineering

Module example: `printf`

Interface is single feature— make formatted string

```
int printf(const char* format, ...);
```

Implementation is complex, must:

- Parse format string
- Convert arguments into strings
- Concatenate conversions into output string
- Print output over uart

`snprintf()`

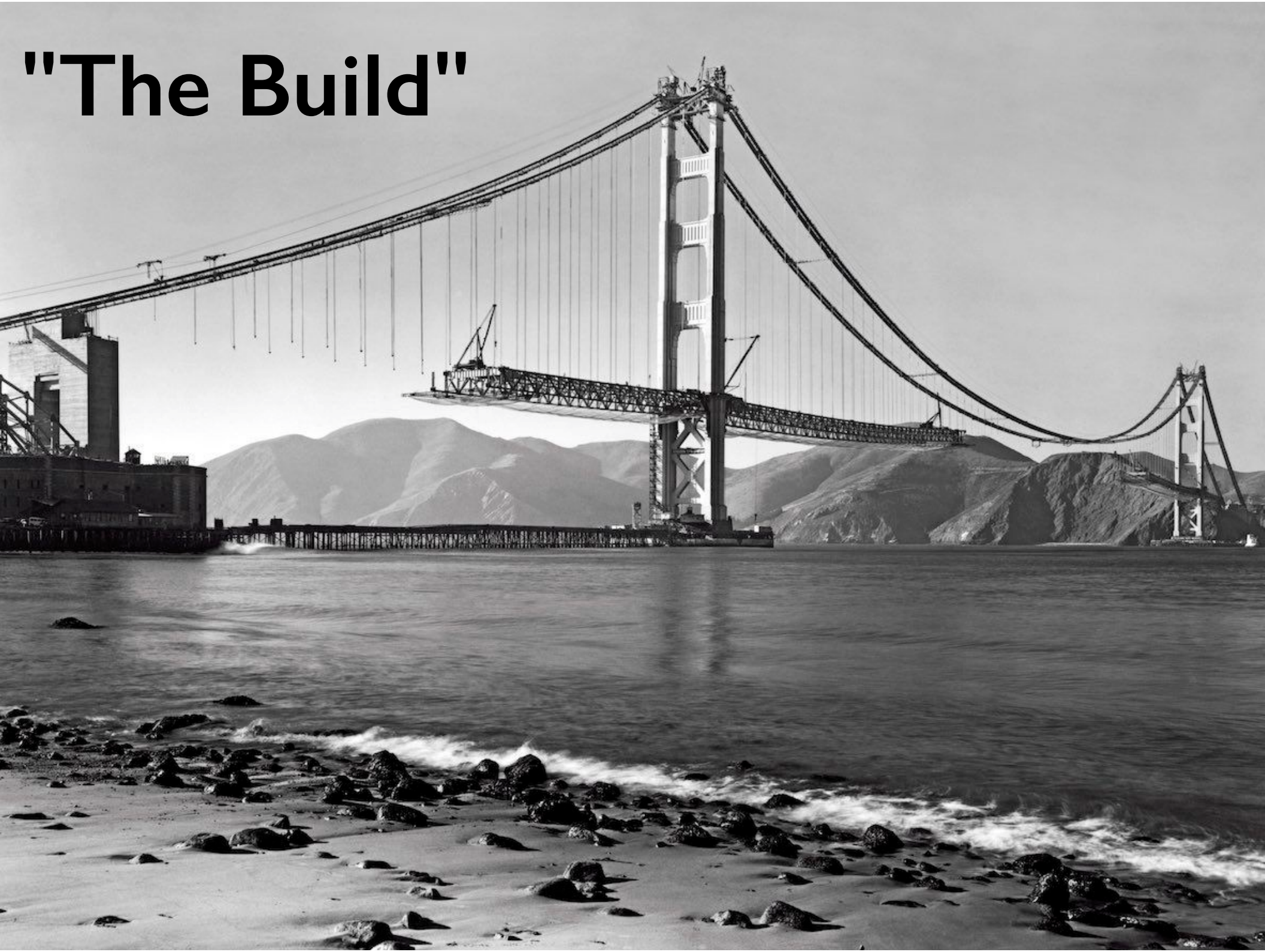
`num_to_string()`

`strlcat()`

`uart_putstring()`

Decompose problem into smaller, simpler tasks, test independently, build up in layers

"The Build"



Lecture code examples

`Link to code folder is on lecture schedule`

`code/clock`

multi-module build (similar to assign2)

`code/clock-libmango`

clock program link with libmango

`code/data`

example of global data variables

Guide to binutils

`as, ld, objcopy, ar`

Tools for building program binaries

`nm, size, strings, objdump`

Tools for examining program binaries

NASA Command Center Artemis II mission

CS107e Command Center: `Makefile`

```
APPLICATION = clock
MY_MODULES = gpio.o timer.o

ARCH    = -march=rv64im_zicsr -mabi=lp64
ASFLAGS = $(ARCH)
CFLAGS  = $(ARCH) -Og -g -Wall -ffreestanding
LDFLAGS = -nostdlib -T memmap.ld
LDLIBS  =

all : $(APPLICATION).bin

%.bin: %.elf
   riscv64-unknown-elf-objcopy $< -O binary $@

%.elf: %.o $(MY_MODULES) cstart.o start.o          LINKER
   riscv64-unknown-elf-ld $(LDFLAGS) $^ $(LDLIBS) -o $@

%.o: %.c                                           COMPILER
   riscv64-unknown-elf-gcc $(CFLAGS) -c $< -o $@

%.o: %.s                                           ASSEMBLER
   riscv64-unknown-elf-as $(ASFLAGS) $< -o $@
```

Build process

PREPROCESSOR: cpp `program.c -> program.i`

Input: C source file

Output: replace **`#include`** and **`#define`** with text expansions

COMPILER: cc `program.i -> program.s`

Input: preprocessed C source file

Output: assembly file

ASSEMBLER: as `program.s -> program.o`

Input: assembly file

Output: machine code (binary-encoded)

LINKER: `ld` `program.o -> program.elf`

Input: multiple .o files, libraries

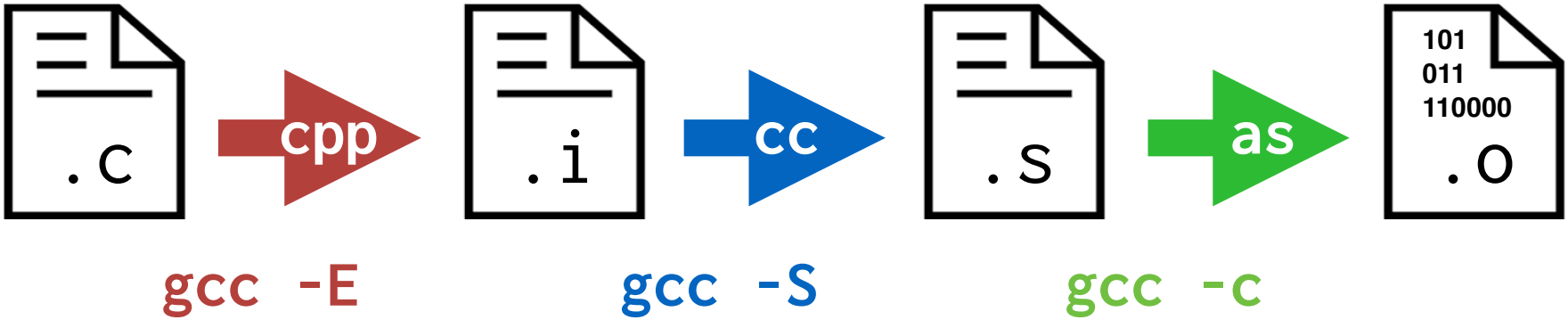
Output: executable in ELF

`objcopy` `program.elf -> program.bin`

Input: ELF executable

Output: raw binary

Compiling a single module

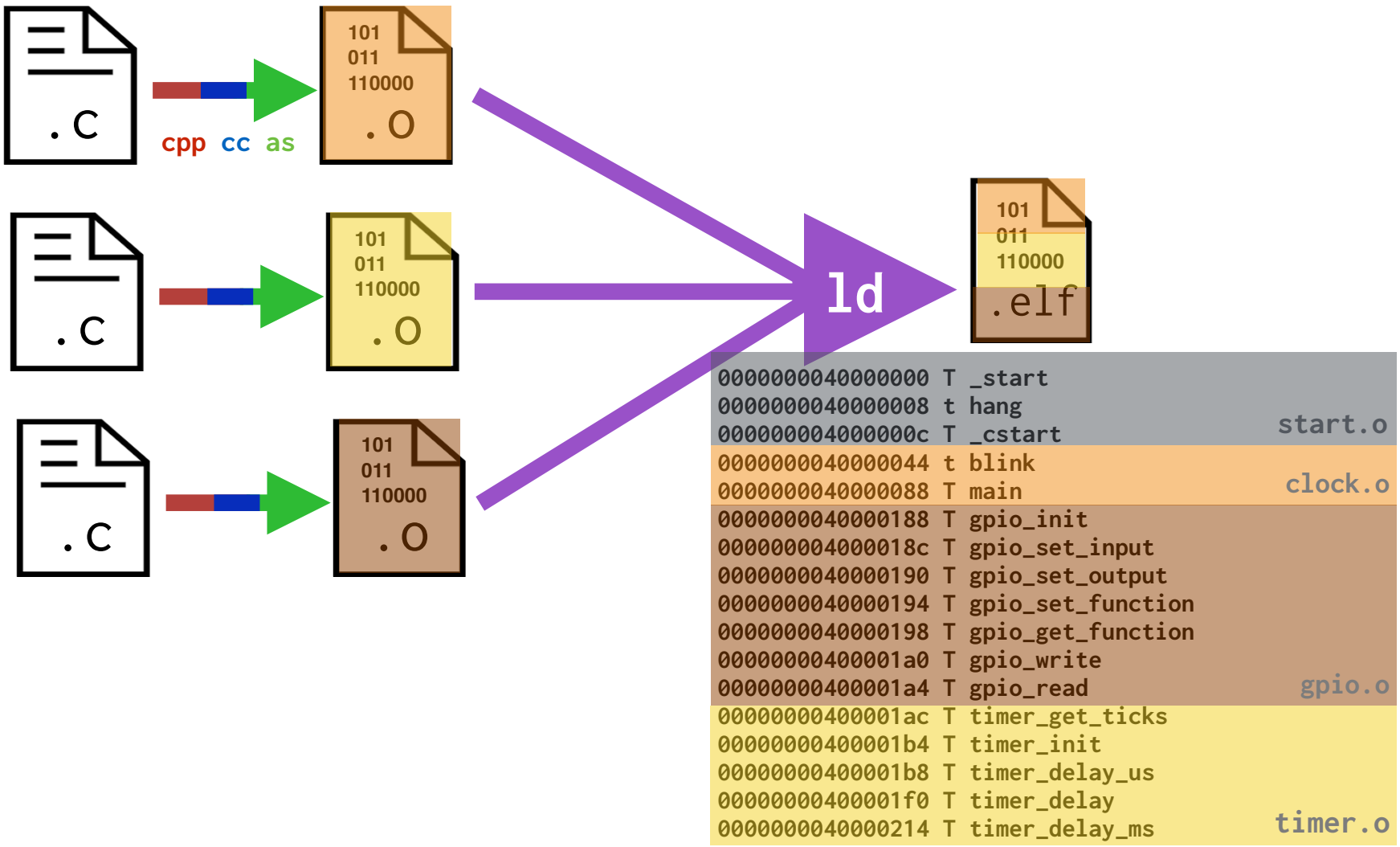


want to see intermediate files?

```
gcc --save-temps
```

Link modules into executable

```
$ riscv64-unknown-elf-ld -nostdlib -T memmap.ld
                          clock.o gpio.o timer.o start.o -o clock.elf
```



What does the linker do?

```
$ riscv64-unknown-elf-ld -nostdlib -T memmap.ld
                        clock.o gpio.o timer.o start.o -o clock.elf
```

Combines one or more object files into executable

Resolve inter-module references

Coalesce like sections

Fix addresses of code and data so program can run

Arrange all into output file in proper order

Q: Why need linker? Why can't compiler do it?

Symbols

C has single global namespace

- Each name can have only one definition
- Need conventions to avoid name clashes
  - e.g. **`gpio_init`** versus **`timer_init`**

Qualifier dictates symbol visibility and link behavior

- **`static`** private to module, not visible outside, no linking needed
- **`extern`** public, visible to other modules, cross-module references need link

Linker only concerned with **extern** symbols

Symbol resolution

Rule: Every symbol referenced must be defined <u>once and exactly once</u>

Linker errors during symbol resolution:

- **Multiply-defined**: two definitions for same symbol
- **Undefined**: needed symbol never defined

Linker resolution process

```
$ riscv64-unknown-elf-ld -nostdlib -T memmap.ld
                         clock.o gpio.o timer.o start.o -o clock.elf
```

Set D (symbols that have been defined)

Set U (symbols that have been referenced, but not yet defined)

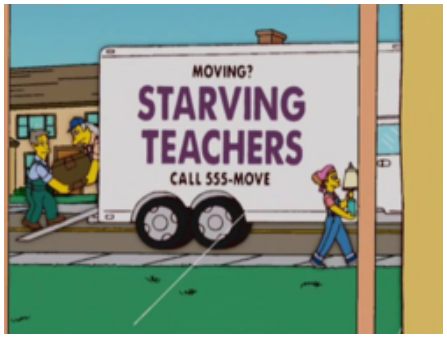
Linker processes .o files in order from left to right

Process each .o updates **Set U** and **Set D**

Continue until all .o processed

If end with **Set U** empty and **Set D** has no duplicates, link successful!

Symbol relocation



In .o file:

- each symbol assigned a temporary address (offset within module)
- reference to symbols within same module is PC-relative

In executable file:

- modules consolidated in order, laid out end to end
- first module base address **`0x40000000`**
- base address of subsequent module adds size of all previous modules
- address of symbol = `module_base + symbol_offset_within_module`

Before link

start.s

```
_start:
    lui     sp,0x50000
    jal     _cstart
hang: j hang
```

```
$ riscv64-unknown-elf-objdump -d start.o
```

```
    0:  lui    sp,0x50000
    4:  jal    0
    8:  j      8 <hang>
```

Target address for `j hang` is 8, encoded as PC-relative offset (in this case, `pc+0`)

Target address for `jal _cstart` is 0 (placeholder!).
File `start.s` doesn't know where `_cstart` is! (Why not?)

Binutils: `objdump -d` shows disassembly

After link

```
$ riscv64-unknown-elf-objdump -d clock.elf
```

```
0000000040000000 <_start>:
    40000000:   lui sp,0x5000
    40000004:   jal 4000000c <_cstart>

0000000040000008 <hang>:
    40000008:   j  40000008 <hang>
```

All addresses have been relocated/finalized
Address of **_cstart** is now known — **4000000c**

The linker worked it out and replaced target placeholder with function's actual address

Handling of global data

Variable declared outside functions has **global** scope and **program** lifetime
(can be extern/static, read-only, initialized/not)

```
                        // uninitialized
int gNum;
static int sgNum;

                        // initialized
int iNum = 0x1;
static int siNum = 0x22;

                        // const
const int cNum = 0x333;
static const int scNum = 0x4444;
```

Binutils: `size`, `nm`

```
$ riscv64-unknown-elf-size data.o
   text    data     bss     dec     hex filename
    377       8       8     393     189 data.o

$ riscv64-unknown-elf-nm -S data.o
0000000000000004 0000000000000004 R cNum
0000000000000004 0000000000000004 B gNum
0000000000000004 0000000000000004 D iNum
000000000000002c 00000000000000d4 T main
                 U printf
0000000000000000 0000000000000004 r scNum
0000000000000000 0000000000000004 b sgNum
0000000000000000 000000000000002c t show_var
0000000000000000 0000000000000004 d siNum
                 U uart_init
```

<u>LEGEND</u>

`T/t` = text (code)

`D/d` = read-write data

`R/r` = read-only data (**`const`**)

`B/b` = bss (*Block Started by Symbol*)

lowercase **`static`** vs uppercase **`extern`**

`U` = undefined

Libraries

An archive .a is just a collection of modules (.o files)

Linker looks in library to find definitions for symbols in **Set U** (undefined). When symbol is found, entire module is linked in.

If linking this module results in new undefined symbols, they are added to **Set U**. Process repeats until no additional undefined symbols, move on to next library

Building with libmango

```
ARCH    = -march=rv64im -mabi=lp64
CFLAGS  = $(ARCH) -Og -I$$CS107E/include -Wall -ffreestanding
LDFLAGS = -nostdlib -L$$CS107E/lib -T memmap.ld
LDLIBS  = -lmango

%.bin: %.elf
   riscv64-unknown-elf-objcopy $< -O binary $@

%.elf: %.o
   riscv64-unknown-elf-ld $(LDFLAGS) $^ $(LDLIBS) -o $@

%.o: %.c
   riscv64-unknown-elf-gcc $(CFLAGS) -c $< -o $@
```

In Makefile for assigns 3 - 7,
`MY_MODULE_SOURCES` controls which of your modules to link,
otherwise will use from libmango

Diagnosing build errors

Which tool detects/reports… ?

mismatched type

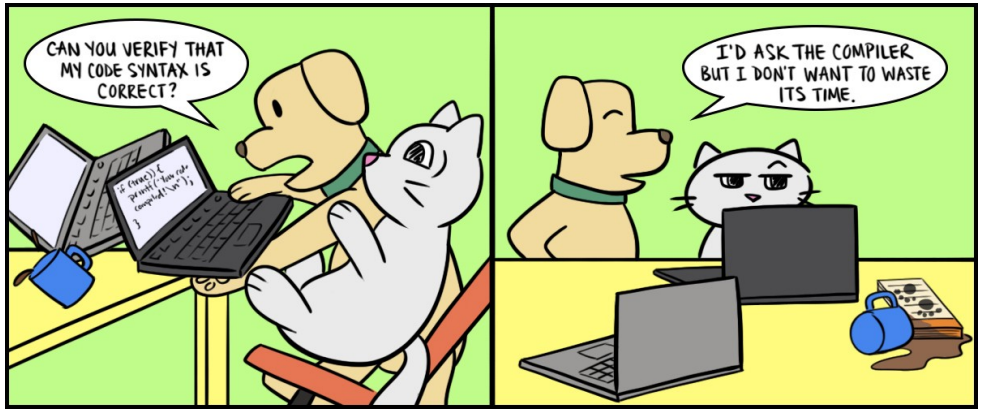
function call with wrong arguments

forgot #include

forgot to link with library

use variable uninitialized

two functions with same name



Modules and build process

Decompose into modules is skill to cultivate

- Separate system into coherent modules, minimize dependencies
- Clean abstractions, tidy interface
- Develop and test independently

Automate the build process

- Makefile is your command center
  - **Fast**: re-compile only what has changed
  - **Reliable**: track dependencies between modules
    - If file F changes, recompile everything that depends on F

Program start sequence

How is a program loaded into memory?

What must happens to start executing a program?

What is known about state of registers or contents of memory?

Does execution actually start at `main()`

👀 **Read our files!** 👀

`$CS107E/src/start.s`

`$CS107E/src/cstart.c`

`$CS107E/lib/memmap.ld`

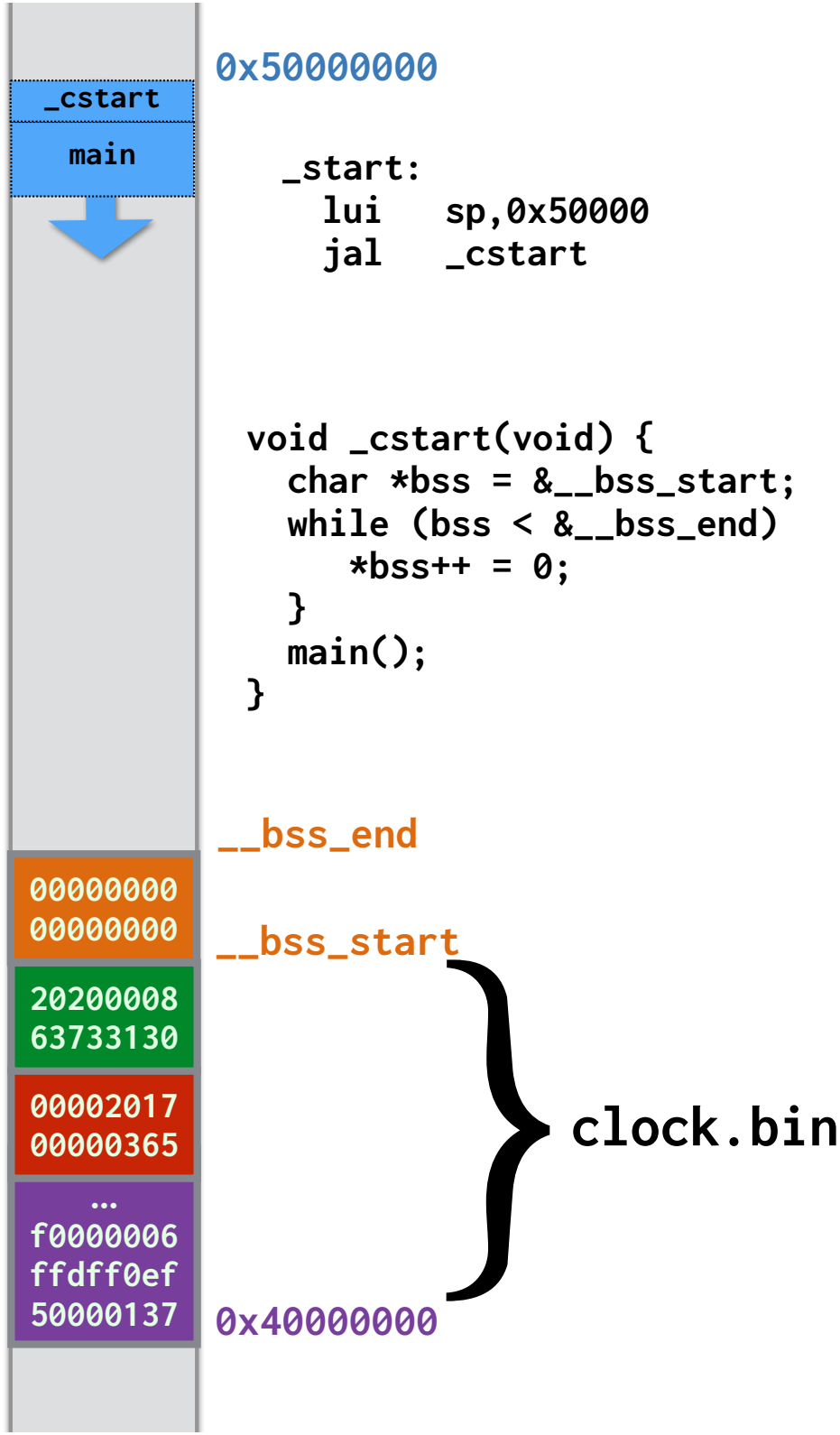
`memmap.ld` (linker script)

```
SECTIONS
{
    .text 0x40000000  :  { *(.text.start) *(.text*) }
    .rodata           :  { *(.rodata*) }
    .data             :  { *(.sdata*) }
    __bss_start       = .;
    .bss              :  { *(.bss*) }
    __bss_end         = .;
}
```

What is special about `.text.start` and why must it go first?

What is the significance of `0x40000000`?

```
SECTIONS
{
  .text 0x40000000 :{ *(.text.start)
                      *(.text*)}
  .rodata :          { *(.rodata*) }
  .data :            { *(.data*) }
  __bss_start        = .;
  .bss :             { *(.bss*) }
  __bss_end          = . ;
}
```



(zeroed data) .bss

(initialized data) .data

(read-only data) .rodata

.text

```
$ xfel write 0x40000000 clock.bin
$ xfel exec  0x40000000
```